

Latest developments in $\text{\LaTeX}$

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Introduction



- For the first two decades, the kernel remained largely stable, with minimal bug fixes.
- Additional features were added primarily through (standard) standalone packages: graphics, babel, color, fontenc, inputenc, calc, longtable, etc.
- But the world keeps turning, with advances and new requirements in Unicode, fonts, accessibility in PDF documents, etc.
- In 2015 development changed significantly: now the core system is often directly updated.
- The purpose of this talk is to highlight *some* of the latest developments.



1994-06-01	1999-06-01	2015-01-01	2020-10-01
1994-12-01	1999-12-01	2015-10-01	2021-06-01
1995-06-01	2000-06-01	2016-02-01	2021-11-15
1995-12-01	2001-06-01	2016-03-01	2022-06-01
1996-06-01	2003-12-01	2017-01-01	2022-11-01
1996-12-01	2005-12-01	2017-04-15	2023-06-01
1997-06-01	2007-12-01	2018-04-01	2023-11-01
1997-12-01	2009-09-24	2018-12-01	2024-06-01
1998-06-01	2011-06-27	2019-10-01	2024-11-01
1998-12-01	2014-05-01	2020-02-02	2025-06-01



- Most of changes are extensions, with new features and new tools.
- Or different ways (hopefully for the better) to do some things.
- With these changes, there is a lot of outdated information out there.
- Be aware AI chats are often not up to date either!
- Compatibility is still a concern, but sometimes some incompatibilities are inevitable.
- Since the code is being continuously revised, some bugs may be introduced. You can help by reporting them.

First changes



- One of the earliest significant changes.
- Unicode has seized the world.
- Most editors save files in UTF-8 by default.
- Unicode UTF-8 encompasses all scripts in all systems.
- The input encoding mess has been solved.



Formerly

```
\documentclass{article}
```

```
\usepackage[T1]{fontenc}
```

```
\usepackage[utf8]{inputenc}
```

```
\usepackage[portuguese]{babel}
```

```
\begin{document}
```

De grão em grão, a galinha enche o papo.

```
\end{document}
```



Currently (we will return to this later)

```
\documentclass{article}
```

```
\usepackage[T1]{fontenc}
```

```
\usepackage[portuguese]{babel}
```

```
\begin{document}
```

De grão em grão, a galinha enche o papo.

```
\end{document}
```



- Originally devised as a sort of replacement for L^AT_EX.
- Now, it has become a programming layer.
- Formerly has to be loaded explicitly.
- Since 2020 it's part of the L^AT_EX core and directly available.
- Including some tools to define commands formerly available only as a package (xparse, xtemplate, ...).
- Other tools provided by the L^AT_EX3 layer are floating-point calculations, basic regular expressions, data list manipulation, dictionary-like structures (key/value), etc.
- Some of the developments I'll mention are based on the L^AT_EX3 layer.
- Mainly addressed to developers.



- Improving $\text{\LaTeX}$ with new features often requires redefining internal macros.
- Several packages may modify the same macro. For example, the internal macro `\@footnote` is redefined by 7 packages.
- This leads to incompatibility between packages.
- Users may also want to modify some hardcoded behaviors.
- There were tricks, like those based on `\let`, which I won't explain because they are (usually) not needed anymore.
- Also with the package `etoolbox`.
- The consequence: frustration in package developers and users.



- A hook is a place in the code where new code can be inserted.
- There are currently many hooks.
- It can be also used to patch commands and environments in a safe way.

► Hooks and tabular (double click)



- Start the long-term Tagging PDF project.
- This deserves special attention.



- Formerly based on the T_EX primitives `\uppercase` and `\lowercase`.
- But Unicode text showed its limitations.
- There were also some annoying results. With `\MakeUppercase{Uppercase with math  $x^2 = y$ }`, you got:

UPPERCASE WITH MATH $X^2 = Y$



- A completely new way to deal with them has been devised.
- Another example of a improved feature thanks to the $\text{\LaTeX}$ 3 layer.
- Deals with strings more cleverly. With `\MakeUppercase{Uppercase with math  $x^2 = y$ }`, you get:

UPPERCASE WITH MATH $x^2 = y$



- They can be localized. For example, in Latin:

‘lupus’ → ‘LVPVS’

- And also

`\MakeTitlecase[words=all]{Titlecase with
math $x^2 = y$ }`:

Titlecase With Math $x^2 = y$



- Lua is a small programming language with a simple syntax created by Brazilian computer scientist Roberto Ierusalimschy.
- LuaT_EX combines Lua with pdfT_EX (and some features from Omega, by Yannis Haralambous and John Plaice). LuaL^AT_EX is LuaT_EX + L^AT_EX.
- pdfL^AT_EX still supported.
- On the other hand XeL^AT_EX is not recommended: it's unmaintained and its support for PDF is limited.



- It supports modern font technologies (OpenType, TrueType).
- With its integration with T_EX, you have access to many of the internals of the T_EX engine.
- Actually, you don't need to learn Lua. Only if you need those extra features.
- Proper Tagged PDF generation is an example of a feature requiring LuaT_EX.
- Almost anything that can be done with XeT_EX can be done with LuaT_EX (very likely everything). And LuaT_EX can do a lot more.



Currently (again)

```
\documentclass{article}

\usepackage[portuguese]{babel}

\begin{document}

De grão em grão, a galinha enche o papo.

\end{document}
```

Tagged PDF



They can contain audio, video, JavaScript. They can be interactive. They can contain also additional information or be tailored for specific purposes.

- **PDF/A** For long-terms archiving.
- **PDF/E** For Engineering documents.
- **PDF/X** For preprinting data eXchange.
- **PDF/UA** For Universal Accessibility. Tagging the PDF is one important component of this.

(There are several versions of each type.)



- Instead of explaining it, let's show it.
▶ [tagpdf manual](#)
- How to see the tree? PDF-XChange (free), Foxit (free), Acrobat Pro (not so free).
- A little tool is provided: show-pdf-tags.
- More examples: ▶ [Bible](#) ▶ [Max und Moritz](#)
- And even more: ▶ [LaTeX Tagging Project](#)



- **Automatic** and **easy** to use, of course!
- To activate the PDF tagging features (a few more options can be also required).

```
\DocumentMetadata{tagging=on}
```

- **Must** be used even **before** \documentclass.
- Even without tagging, it may add some features. The font encoding is switched to T1 by default (2024), and the PDF version is set to 2.0 (2025).
- ▶ Tagging 1 (double click)
- A larger example: ▶ Larger example .



On Tagged PDF:

From “PDF Techniques for WCAG 2.0” (W3C)

It defines a set of standard structure types and attributes that allow page content (text, graphics, and images) to be extracted and reused for other purposes.

In other words, the information is presented not only visually, but also in a structured way.



- **Text and graphic extractions.** So that they can be copied and pasted into other applications.
- **Content flow.** The logical order is known, thus allowing a different layout from that assumed for the original one, and ensures a screen reader moves through the document in the correct sequence.
- **Searching and indexing.** It should be possible to analyze its contents, including spell-checking.
- **Format conversion.** To HTML, XML, etc., preserving the structure and the basic styling information.
- **Accessibility.** Last but not least, making content accessible to people who rely on assistive technology (AT), including reading aloud or generating braille.



There are two standards: PDF/UA-1 (PDF v. 1.7) and PDF/UA-2 (PDF v. 2.0).

- Improve accessibility of PDF documents produced by $\text{\LaTeX}$.
- Improve the ability to better understand the document content. ▶ Tagging 1 (double click)
- Push PDF 2.0. Better for accessibility. Required in math, but currently poorly supported.
- Push improvements in viewers and tools.
- A weird reading without tagging: ▶ Dots. Video taken from “Listening to untagged and tagged PDF”.



HTML

- Structured language.
- All viewer understand the structure.
- Lots of books and guidelines.
- Behaviour of assistive technologies (AT) tools is well understood.

PDF

- Page description language.
- Structure must be added on top, but viewer can (do) ignore it.
- Scarcely documented (very technical; designed for GUIs).
- Behaviour of assistive technologies tools (AT) is still unclear.



HTML

- Exact rendering depends on the browser or viewer.
- Except for the text, everything is “external”.
- Difficult to save and use offline (sometimes just impossible).
- Scripts, cookies and session ID dependency.
- Difficult to forward and distribute.
- Difficult to store.

PDF

- Faithful visual representation.
- A single self-contained file: it can even include movies.
- Easy offline use.
- Just double-click and everything is there.
- Easy to forward and distribute.
- Long-term storage (PDF/A).



- Bookmarks for overall structure: navigation.
- Tags for fine-grained structure: comprehension (humans and machines).



- The “Before” (Lamport’s book):
 - Standard classes.
 - Standard commands and environments.
 - Restricted set of packages.
 - hyperref
- The “Now”:
- [▶ Tagging status](#)
- The “After”:
 - Work still in progress.
 - Current work: math.
 - “De grão em grão, a galinha enche o papo.”



- Compatibility with existing packages and even the $\text{\LaTeX}$ kernel.
- Because some macros and environments had to be redefined: `verbatim`, `\item`, `\newtheorem`, etc.
- Adapting existing packages and providing alternatives, including unsupported ones.
- Replacement code in `latex-lab`. And also the option `firstaid`.
- Validating can be also an issue, because some validators are outdated and reject valid Tagged PDF 2.0 files.



- PDF/UA-2 (Universal Accessibility) in PDF 2.0 (ISO 14289-2:2024).
- WTPDF (Well Tagged PDF) for Accessibility and Reuse in PDF 2.0 (free).

Both standards are freely available via pdfa.org.

Other changes



- The core $\text{L}^{\text{A}}\text{T}_{\text{E}}\text{X}$ package for localization and multilingual documents.
- Tailored mainly for $\text{LuaL}^{\text{A}}\text{T}_{\text{E}}\text{X}$, and $\text{pdfL}^{\text{A}}\text{T}_{\text{E}}\text{X}$, but also $\text{XeL}^{\text{A}}\text{T}_{\text{E}}\text{X}$.
- Starting with 2017 a new core, with a modern reworked way to deal with languages and locales has been devised, based on descriptive `ini` files. `babel-pt.ini`
- Based on latest advances in Unicode (including the Unicode CLDR), the W3C (where applicable) and other technologies for better compatibility and consistency.
- It supports ~300 languages (with various levels of coverage) across ~45 scripts, including complex and RTL ones like Arabic, Chinese, Tibetan and Devanagari, and ancient languages like Phoenician, Ugaritic and Ancient Egyptian.



- Integration with OpenType technologies in Lua $\text{\LaTeX}$ and Xe $\text{\LaTeX}$, with a high-level interface to fontspec.
- Lazy loading of languages and fonts. Only if required.
- Multilingual strings. ► Multilingual strings
- Tools for tailoring: `\setlocalecaption`.

Do

```
\setlocalecaption{brazilian}{contents}{Índice}
```

Don't

```
\addto\captionsbrazilian{%  
  \def\contentsname{Índice}}
```



- Inspired by Omega transformation processes.
- It can provide language dependent tools based on regex-like rules.
- Thanks to the access to the $\text{T}_{\text{E}}\text{X}$ engine with Lua $\text{\LaTeX}$.
- Example for Portuguese explicit hyphen. ▶ Hyphen (double click)
- (Actually, the same effect can be achieved with the built-in parameter `\postexhyphenchar`.)
- More examples: ▶ Transforms and ▶ Arabic.

A few miscellaneous improvements



- `\llap` and `\rlap` have existed for ages:

`\llap`, `\rlap`

Before `\rlap{-}` after

Before ~~after~~

Before `\llap{-}` after

Before~~after~~

- But not for a centered overlapping. Now we have `\clap`.

`\clap`

Before `\clap{-}` after

Before~~after~~



- Formerly only values relative to `\unitlength`.
- Now absolute values (i.e., with the unit) are also allowed.

picture

```
Some text \begin{picture}(1.5em,0)
\color{red}
\put(1.5em, .5ex){\vector(-1,0){1.2em}}
\end{picture}.
```

Some text .

- By the way: with `picture` the `pict2e` package is recommended.



- Useful in didactical works.
- But far from trivial.

\mathcolor

```
\[ \mathcolor{blue}{\sum}_{i=1}^n x_i  
\mathcolor{blue}{ = \left\{ \right. \frac{1}{2}  
\mathcolor{blue}{\right\}^{\alpha+\beta} \]
```

$$\sum_{i=1}^n x_i = \left\{ \frac{1}{2} \right\}^{\alpha+\beta}$$



L^AT_EX

- “The L^AT_EX Companion”, 3rd ed., 2023.
- ltnews – The latest L^AT_EX news.
- Some TeX Developments (Joseph Wright). Also RSS.

Tagged PDF

- L^AT_EX Tagging Project.

Babel

- Babel site.